Applied Mechanics and Materials Vols. 105-107 (2012) pp 1970-1974
Online available since 2011/Sep/27 at www.scientific.net
© (2012) Trans Tech Publications, Switzerland
doi:10.4028/www.scientific.net/AMM.105-107.1970

Study on Uncertainty of Gas Monitoring Dynamic Calibration System

Li De Fang[a], Xiu Ming Xiang[b], Xiao Ting Li[c], Li Li Pang[d], Xiao Jie Wang[e], Ran Liu

The college of Quality and Technical Supervision, Hebei University, Baoding 071002, China

[a]fanglide@sina.com, [b]xiangxiuming@126.com,[c] lxt@hbu.edu.cn, [e]pangnan0315@163.com

Keywords: Gas monitoring, Dynamic calibration system, Measurement uncertainty, Mass flow controller

Abstract. With the rapid development of industry and transportation, air pollution is worsening, therefore, monitoring of air pollution components is more and more heeded. In this study, based on the measurement model and the composition of the air quality monitoring system on line, the mathematical model of the measurement system value transmission was analyzed, and the uncertainty components were calculated respectively, a conclusion that the main factor of the overall system uncertainty is the uncertainty of system itself in the existing air quality monitoring system was drawn, so measurement uncertainty of the calibration system was focused on the research. The effects of the Zero-gas, calibration gases and gas mass flow controller on the uncertainty of the calibration system was experimented and analyzed, and measurement uncertainty of the dynamic calibration system was evaluated.

Introduction

Air pollution is from pollutants emitted by stationary sources and mobile sources, and the spread of pollution emissions is a function of emissions and practice space, so it is a large-scale of air quality monitoring, the air quality monitoring has a strongly influence by season, climate, topography, surface features, etc. and changes over time [1,2]. Air quality monitoring system consists of a central station and a number of sub stations. The central station is the heart of the whole system. Sub stations are equipped with automatic sampling and pretreatment system, a variety of gas analysis and monitoring instrumentation, zero gas generator and the dynamic calibrator [3-7].

Monitoring of meteorological parameters include temperature, humidity, wind speed, etc., detection of pollutants include SO_2, NO_2, NO, CO, H_2S, particulate, total of hydrocarbons, CH_4, Non-methane hydrocarbons etc. [8-13]. Therefore, the air quality on-line continuous monitoring system is a multitasking and multi-module system, the system structure shown in Fig. 1.

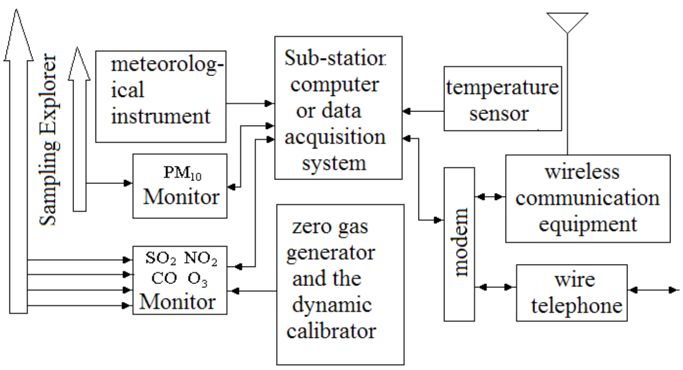

Fig. 1 Continuous ambient air quality monitoring system flow chart

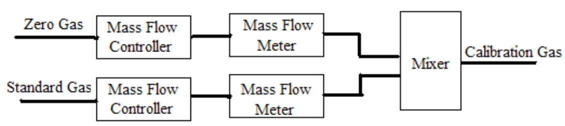

Fig. 2 Dynamic self-calibration unit quantity transmission

It can be seen from Fig. 1, the composition of monitoring system, the gas monitor's measurement uncertainty is affected by many parameters, including the external and internal parameters, and the external parameters are mainly various meteorological parameters, the internal parameters include

All rights reserved. No part of contents of this paper may be reproduced or transmitted in any form or by any means without the written permission of TTP, www.ttp.net. (ID: 140.116.209.14, National Cheng Kung University, Tainan, Taiwan-23/02/14,16:53:25)

system structure, measurement model, the quantity transmission and tractability. This section focuses on the internal parameters. the measurement uncertainty of system data acquisition and communication can increase the median of acquisition card and processor, data acquisition speed is improved, according to micro-error principle, the measurement uncertainty can be ignored, the class B measurement uncertainty of gas monitor mainly comes from dynamic calibrator, so the uncertainty of the dynamic calibrator is a key component of system uncertainty.

Dynamic Calibration System Uncertainty Analysis

Dynamic calibration system's main role is to dynamically calibrate the gas analyzer, including zero gas generators, the standard gas, zero gas and standard gas mass flow control systems, hybrid systems and other components. It is shown in Fig. 2.

It can be seen, the mixed gas that the calibration apparatus output to gas analyzer, which concentrations is determined by,

$$n = \frac{n_1 q_1 + n_2 q_2}{q_1 + q_2} \tag{1}$$

n is the volume fraction of mixed standard gas, n_1 is the volume fraction of diluted standard gas in zero gas, n_2 is the volume fraction of standard gas, q_1 is zero gas flow, q_2 is standard air flow.

The uncertainty mixed standard gas concentration can be expressed as combined standard uncertainty,

$$u_n^2 = \left(\frac{\partial n}{\partial n_1} u_{n_1}\right)^2 + \left(\frac{\partial n}{\partial q_1} u_{q_1}\right)^2 + \left(\frac{\partial n}{\partial n_2} u_{n_2}\right)^2 + \left(\frac{\partial n}{\partial q_2} u_{q_2}\right) \tag{2}$$

The sensitivity coefficients are, $\frac{\partial n}{\partial n_1} = \frac{q_1}{q_1 + q_2}$, $\frac{\partial n}{\partial n_2} = \frac{q_2}{q_1 + q_2}$, $\frac{\partial n}{\partial q_1} = \frac{q_2(n_1 - n_2)}{(q_1 + q_2)^2}$, $\frac{\partial n}{\partial q_2} = \frac{q_1(n_2 - n_1)}{(q_1 + q_2)^2}$

u_n is the uncertainty of Mixed standard gas volume fraction, u_{n_1} is the uncertainty of zero gas volume fraction, u_{n_2} is the uncertainty of standard gas volume fraction, u_{q_1} is the uncertainty of zero gas flow, u_{q_2} is the uncertainty of standard gas flow. With a relative standard uncertainty is expressed as,

$$\left(\frac{u_n}{n}\right)^2 = \left(C_1 \frac{u_{n_1}}{n_1}\right)^2 + \left(C_2 \frac{u_{q_1}}{q_1}\right)^2 + \left(C_3 \frac{u_{n_2}}{n_2}\right)^2 + \left(C_4 \frac{u_{q_2}}{q_2}\right)^2 \tag{3}$$

Where, $C_1 = \frac{q_1 n_1}{n_1 q_1 + n_2 q_2}$, $C_2 = \frac{q_1 q_2 (n_1 - n_2)}{(q_1 + q_2)(n_1 q_1 + n_2 q_2)}$, $C_3 = \frac{q_2 n_2}{n_1 q_1 + n_2 q_2}$, $C_4 = \frac{q_1 q_2 (n_2 - n_1)}{(q_1 + q_2)(n_1 q_1 + n_2 q_2)}$

Experimental Data and Analysis

Based on the actual system requirements, taking the total flow rate 650 ml/min and zero gas is high-purity nitrogen or dried and purgative air. Specific requirements are, high-purity nitrogen purity is better than 99.999%, for the dried and purgative air, the dew point is <-20℃, $SO_2<5\times10^{-9}$, $NO<5\times10^{-9}$, $NO_2<5\times10^{-9}$, $O_3<5\times10^{-9}$, $HC/CO<5\times10^{-9}$. Standard gas is the national primary standard SO_2 by state approved, or configured by permeation tube or standard gas generating device. The uncertainty requirement is no more than ±1% when gas volume fraction is greater than 10×10^{-6}, and no more than ±2% when gas volume fraction is less than 10×10^{-6}. To purify the air as the zero gas, volume fraction of being diluted standard gas is less than 5×10^{-9}, 4×10-9 is selected. According to the Eq.1, configuring standard gas, which volume fraction is 50ppb, 100ppb, 200ppb, 300ppb, and 400ppb. Then the calculated sensitivity coefficients by Eq.3 are in the following table.

Table 1 Calculation parameters

zero gas volume fraction	standard gas volume fraction	zero gas flow [ml/min]	standard gas flow [ml/min]
0.000000004	0.00001	647.0088035	2.991196479
0.000000004	0.00001	643.757503	6.242496999
0.000000004	0.00001	637.254902	12.74509804
0.000000004	0.00001	630.7523009	19.24769908
0.000000004	0.00001	624.2496999	25.75030012

Table 2 Sensitivity coefficients

C_1	C_2	C_3	C_4
0.079631852741	-0.915766306523	0.920368147259	0.915766306523
0.039615846339	-0.950780312125	0.960384153661	0.950780312125
0.019607843137	-0.960784313725	0.980392156863	0.960784313725
0.012938508737	-0.957449646525	0.987061491263	0.957449646525
0.009603841537	-0.950780312125	0.990396158463	0.950780312125

As can be seen in Fig. 3, C_1 is smaller than other sensitivity coefficients. Its value is decreasing with Standard gas volume fraction increasing from table 2, so C_1 Contribution to the total uncertainty which is also reduced, can be ignored the first component of uncertainty. The absolute value of other three sensitivity coefficients is almost equal size.

The Impact of Component Uncertainty to Overall Uncertainty

Considering the impact of standard gas volume fraction uncertainty to mixed standard gas uncertainty, according to Eq.1, in case of the other parameters is constant, and it is change of the standard gas volume fraction uncertainty, then,

$$\frac{u_n}{n} = \frac{q_1 n_1}{n_1 q_1 + n_2 q_2} \frac{u_{n_1}}{n_1} \tag{4}$$

Similarly available, $\frac{u_n}{n} = \frac{q_1 q_2 (n_1 - n_2)}{(q_1 + q_2)(n_1 q_1 + n_2 q_2)} \frac{u_{q_1}}{q_1}$, $\frac{u_n}{n} = \frac{q_2 n_2}{n_1 q_1 + n_2 q_2} \frac{u_{n_2}}{n_2}$, $\frac{u_n}{n} = \frac{q_1 q_2 (n_2 - n_1)}{(q_1 + q_2)(n_1 q_1 + n_2 q_2)} \frac{u_{q_2}}{q_2}$

According to the parameters in Table 1, it can be obtained the impact of different components uncertainty to the overall uncertainty from Eq.4. It is shown in Fig. 4-6.

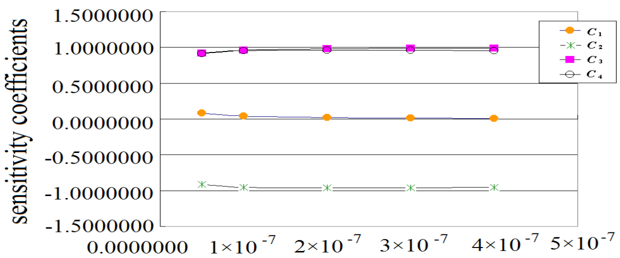

Fig. 3 The relative standard uncertainty sensitivity coefficients

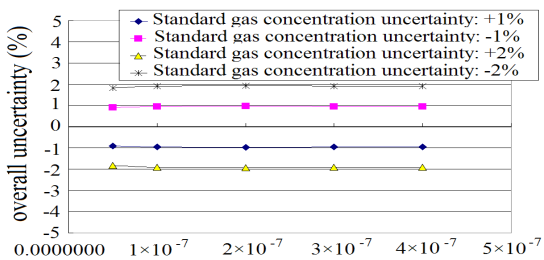

Fig. 4 The impact of standard gas volume fraction uncertainty to overall uncertainty

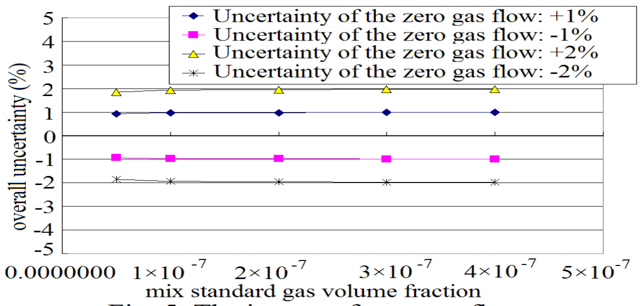

Fig. 5 The impact of zero gas flow uncertainty to overall uncertainty

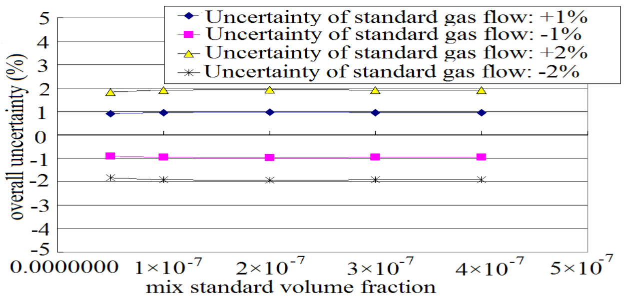

Fig. 6 The impact of standard gas flow uncertainty to overall uncertainty

As can be seen from the above chart, the main factors that impact calibration system uncertainty are the uncertainty of standard gas volume fraction, the uncertainty of the zero gas flow, and standard gas flow uncertainty. It is equivalent influence of the three components.

Conclusion

Considering the impact of three-component, the overall uncertainty is,

$$\sqrt{\left(\frac{u_n}{n}\right)^2}=\sqrt{\left(C_2\frac{u_{q_1}}{q_1}\right)^2+\left(C_3\frac{u_{n_2}}{n_2}\right)^2+\left(C_4\frac{u_{q_2}}{q_2}\right)^2} \tag{5}$$

When the uncertainties of three components are taken at ± 1% and ±2%, respectively the overall uncertainty is ±1.65% and ±3.30%. It can be seen, when the mass flow controller uncertainty is ±1%, and leads to standard gas that the uncertainty is ±1%, the theoretical uncertainty is ± 1.65% of standard gas mixture re-emerged by calibration system, but it is not considered other structural of system and environmental factors. However, in the evaluation of measurement uncertainty to gas analyzer, a lot of literatures are usually considered the standard uncertainty of the gas volume fraction is mixed standard gas uncertainty.

Acknowledgements

This paper is supported by the Natural Science Foundation of China (61074174), Natural Science Foundation of Hebei Province (F2011201040), and Natural Science Foundation of Hebei University (2009-169).

References

[1] X.Y. Yang: *Design of and Experimental Study on Atmospheric Pollutant Monitoring System with UV Spectroscopy* (MS., Tianjin University, China 2005.1). (In Chinese)

[2] B.C. Wu, L. Fei: *Modern Environmental Monitoring Technology*(China Environmental Science Press, China 2005). (In Chinese)

[3] X.J. Hao, K.J. Li and J. Liu: Acta ArmamentarII, Vol. 30 (2009.2) No.2, p.30-33. (In Chinese)

[4] L. Michalski, K. Eckersdorf, J. Kucharski and J. McGhee: *Temperatures measurement*(John Wiley & Sons, UK 2001).

[5] J.J.Yuan, J.Z and J.G.Yang: Measurement and Test Technology, Vol. 36 (2009) No.2, p.31-32. (In Chinese)

[6] L.Huang: *Research on Dynamic Calibration Technology of Transient Surface Temperature Sensor* (MS, North University of China, China 2007). (In Chinese)

[7] C.P.Yang: *Theories and Techniques Studies on Ultra-Violet Fluorescence Detection of Sulfur Dioxide in Air the Parts per Billion Level* (MS., South China Normal University, China 2002) (In Chinese)

[8] W.M.Wen: *Development of an Indoor Toxic Gas Monitoring System*(MS, Xi'an University of Science and Technology, China 2008.4.15). (In Chinese)

[9] W.Q.Qi, Z.G.Sun and G.G.Bian: *Environmental Monitoring New Technology*(Chemical Industry Press, China 2004). (In Chinese)

[10] D.G. Streets, N.Y. Tsai, H. Akimoto and K. Oka: Atmospheric Environment, Vol. 34 (2000) p.4413.

[11] U.S. Environmental Protection Agency: *National Air Quality and Emissions Trends Report*, (U.S. Environmental Protection Agency, U.S. 1993), p.136.

[12] W. Zheng: *Studying of Air Quality Monitoring System Based on FTIR* (MS., Tianjin University, China 2006). (In Chinese)

[13] A. Miketa and L. Schrattenhozer: Energy Policy, Vol. 34 (2006) No.7, p.877.

Vibration, Structural Engineering and Measurement I
10.4028/www.scientific.net/AMM.105-107

Study on Uncertainty of Gas Monitoring Dynamic Calibration System
10.4028/www.scientific.net/AMM.105-107.1970